

Zachary Allen

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WORK EXPERIENCE

Data Scientist

Mar. 2024 – May 2026

Abt Global

Orlando, FL

- Automated AWS Transcribe JSON-to-Word conversion using a serverless MLOps pipeline with EventBridge and Lambda, saving \$80,000 in transcription costs.
- Led creation of LLM-powered entity resolution pipeline with Dagster, Splink, and Claude Sonnet to de-duplicate 150,000+ records, leveraging prompt engineering and API integration to reduce manual processing time from a year to several weeks.
- Optimized statistical model by implementing a Julia-based GLMM with parametric bootstrapping, improving execution time by 55% and memory efficiency by 64% compared to R and Python equivalents.
- Migrated large SAS-based claims ETL to PySpark on Databricks for the Centers for Medicare and Medicaid Services, improving scalability and maintainability for home health quality measures.
- Improved XGBoost model recall by 8% with Optuna hyperparameter tuning as part of an end-to-end ML lifecycle, predicting carbon emission report resubmission probability for the EPA.

Data Scientist Co-op

Jan. – June 2023

Johnson & Johnson, Janssen BioResearch Quality & Compliance

Tampa, FL

- Developed a full-stack Dash Application integrating BERT-based NLP topic modeling to analyze regulatory articles, reducing search time by 33%.
- Performed production feasibility and cost analysis of RAG-based LLM solutions using LangChain for NLP tasks on multiple unstructured PDFs exceeding standard context windows.
- Designed a data pipeline from SharePoint to PostgreSQL with Power Automate and AWS Lambda Functions.
- Created 12 Tableau Dashboards to monitor QA Tool Usage and report on Real-time Transcription Tools.

Data Analyst Intern

June – Aug. 2021

TotalSDS

Tampa, FL

- Leveraged Power BI to develop reports for development productivity, improving Pull Request Lead Time by 40%.
- Completed multiple Data Cleaning/Analysis projects, resulting in a 30% reduction in data entry time and enabling the sales team to make data-driven decisions.

EDUCATION

Georgia Institute of Technology

Jan. 2025 – Present

Masters in Analytics, Computational Data Analytics Track

GPA: 4.0/4.0

University of Florida

Dec. 2023

Bachelors of Science in Computer Science, Minor in Statistics

GPA: 3.88/4.0

PROJECTS

MAISTRO | *Python, Next.js, FastAPI, MongoDB*

Aug. – Dec. 2023

- Designed a chatbot to convert conversation into Spotify song recommendations using GPT-4 and a custom emotion classification model based on Google's 50k observation GoEmotion dataset.
- Implemented front-end with multiple themes and layouts using Next.js, TailwindCSS, and shadcn/ui components.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Julia, TypeScript, C++, Java

Python Libraries: TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Claude API, Plotly Dash, Streamlit, Optuna, HuggingFace Transformers, LangChain

Analytics: PostgreSQL, Databricks, Apache Spark, Dagster, Snowflake, PowerBI, Tableau, Power Automate, Excel

AI/ML: LLM Integration, Prompt Engineering, RAG Pipelines, MLOps, Model Lifecycle Management

Tools: React, Jupyter Notebook, Docker, AWS EventBridge/Lambda, GitHub/Git, LaTeX, PowerPoint